FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

& The Patent Rules, 2003

PROVISIONAL

SPECIFICATION

1. TITLE OF THE INVENTION:

SYSTEM FOR HARVESTING AND UTILIZING ELECTRICAL ENERGY FROM LIGHTNING STRIKES

2. APPLICANT:

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3. PREAMBLE TO THE DESCRIPTION:

The following specification describes the invention and how it will be performed.

BACKGROUND OF THE INVENTION

The present invention relates to a novel and efficient system for harvesting electrical energy from lightning strikes. Harvesting electrical power from natural sources is a field of significant interest, given the increasing global demand for renewable energy.

Lightning, a powerful natural phenomenon, represents an untapped source of electrical energy. Traditional methods of electricity generation, such as dynamo-based systems, convert mechanical energy into electrical energy using various means, including hydraulic forces, steam turbines, wind, and solar power. However, lightning offers an alternative, high-intensity energy source that has not been fully explored due to its unpredictability and the extreme nature of the current involved.

Previous approaches to harnessing energy from lightning have faced challenges in capturing, storing, and safely distributing the enormous energy released during a strike.

In contrast, the present invention provides a novel solution for efficiently capturing, storing, and utilising the vast energy of lightning strikes. The system employs advanced materials, modular design, and robust safety mechanisms to overcome the limitations of prior art and offer a scalable, reliable energy harvesting solution.

SUMMARY OF THE INVENTION

In accordance with the present invention, a novel lightning harvesting system is provided. The system comprises an omnidirectional conductive antenna tower designed to attract and capture lightning strikes, an energy distribution network with ten branches containing high-capacity supercapacitors arranged in parallel, and a control unit configured to manage the storage and release of the captured energy. Additionally, the system includes a grounding mechanism to direct excess current safely to the ground, ensuring the system's safety and reliability.

The antenna tower is strategically designed to maximise the capture of lightning strikes by optimising its height, shape, and surface material. The energy distribution network consists of 10 branches, each containing 1,000 supercapacitors based on Graphene or Activated Carbon, capable of handling high current loads and storing substantial amounts of charge. The control unit plays a crucial role in managing the energy flow, connecting the antenna tower to the supercapacitors during the storage phase and to external loads or grids during the discharge phase. The grounding mechanism ensures that any excess energy not stored is safely dissipated.

The system is designed to be scalable, allowing for deployment in various sizes and regions prone to lightning strikes. This scalability not only enhances the system's adaptability but also its potential impact on energy management.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

FIG. 1 is a schematic diagram of the lightning harvesting system, illustrating the arrangement of the antenna tower, supercapacitors arranged in a box, and grounding mechanism.

FIG. 2 is a schematic diagram of the energy distribution network, showing the ten branches of supercapacitors connected in parallel.

FIG. 3 is a schematic diagram of a high-capacity Supercapacitor based on Graphene and/or Activated Carbon.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS OF THE INVENTION

The following descriptions are exemplary embodiments only and are not intended to limit the invention's scope, applicability or configuration in any way. Instead, the following description provides a convenient illustration for implementing exemplary embodiments of the invention. Various changes to the described embodiments may be made in the function and arrangement of the elements described without departing from the scope of the invention as outlined in the appended claims.

The invention's preferred embodiment is depicted in the combination of FIG. 1 and FIG. 2. The lightning harvesting system comprises several key components:

An embodiment of the invention is noted in the drawings by reference character 100. A preferred embodiment of Invention 100 is depicted in FIG. 1, a schematic view. The lightning harvesting apparatus 100 includes, as one of its primary elements, an omnidirectional conductive antenna tower 110. The antenna tower 110 is constructed from a high-conductivity material, such as copper or an advanced alloy. It is designed to maximise its ability to capture lightning strikes from any direction.

The antenna tower 110 is connected to an energy distribution network 120, configured to distribute the captured energy into multiple branches. The energy distribution network 120 consists of 10 branches, each containing a series of supercapacitors 122 arranged in parallel. Based on graphene or activated carbon, these supercapacitors are designed to handle high currents and store significant charge. The parallel configuration ensures that the energy is evenly distributed across the network, preventing overload and optimising storage.

A control unit 130 is integrated into the system, which manages energy storage and release. The control unit 130 connects the antenna tower 110 to the supercapacitors during the energy storage phase, allowing them to absorb the energy from the lightning strike. During the discharge phase, the control unit 130 connects the supercapacitors to external loads or the grid, enabling a controlled energy release.

A Gas Discharge Tube (GDT) 140 is strategically placed at the end of each branch within the energy distribution network as a grounding mechanism to handle excess current that exceeds the storage capacity of the supercapacitors. This design ensures that the maximum energy is stored before any surplus is safely directed to the ground. When the supercapacitors reach their maximum charge or if there is an unexpected surge, the GDT, acting as the grounding mechanism, activates to discharge any excess voltage, thereby preventing potential damage to the capacitors and other components. This arrangement effectively manages peak lightning currents, maintaining system stability and protecting the overall integrity and longevity of the lightning harvesting apparatus.

The system is designed to be scalable, allowing for deployment in various sizes and regions prone to lightning strikes. This scalability means the system can be adapted to meet different energy needs and lightning frequencies. It can be integrated into regional energy grids, providing supplemental power during peak demand periods or outages and enhancing grid stability and resilience.

The principle of operation of the present invention involves the attraction of lightning strikes by the antenna tower 110, which captures the electrical energy and directs it into the energy distribution network 120. The control unit 130 ensures the energy is stored efficiently in the supercapacitors and safely released when needed. The grounding mechanism 140 prevents system overload, ensuring reliable operation even during extreme lightning events.

Referring to FIG. 1, a front view, these drawings clearly show that, with an assembly of the above-described components, the lightning harvesting system is configured to capture and store electrical energy from lightning strikes efficiently. The omnidirectional design of the antenna tower 110, coupled with the advanced energy storage capabilities of the supercapacitors 122, ensures that the system can harness the immense energy potential of lightning, converting it into a manageable power supply for practical use.

The overvoltage protection component 150 is critical in safeguarding the system from extreme voltages, providing a fail-safe mechanism that prevents damage to the system components. The grounding mechanism 140 and control unit 130 work together to maintain the stability and reliability of the entire system, ensuring continuous power supply even in challenging environmental conditions.

In operation, the lightning harvesting apparatus 100 first attracts a lightning strike, channelling the energy through the antenna tower 110 and into the energy distribution network 120. The energy is then stored in the supercapacitors 122 and held until needed. The control unit 130 manages the energy flow, ensuring efficient storage and safe release while grounding mechanism 140 protects the system from overload.

This innovative system provides a robust method for harnessing and storing lightning energy, offering a significant advancement in energy storage technology with broad applications in areas prone to lightning activity.

The operation of the lightning harvesting system is initiated when a lightning strike is detected by the omnidirectional conductive antenna tower (reference character 100). The antenna tower, constructed from a high-conductivity material, captures the electrical energy from the lightning strike and channels it into the energy distribution network (reference character 200).

Energy Capture and Initial Protection:

Upon the occurrence of a lightning strike, the captured electrical energy is immediately directed through the input line towards the energy distribution network. The input line is equipped with an overvoltage protection circuit to protect the system from the extremely high voltages and currents typically associated with lightning strikes. This circuit includes a series of protective components, such as diodes and resistors, that initially regulate the incoming current and voltage, ensuring that only safe energy levels enter the distribution network.

Energy Distribution and Storage:

Once regulated, the energy enters the energy distribution network, composed of multiple branches, each containing a parallel array of supercapacitors (reference character 210). These supercapacitors, based on materials such as graphene or activated carbon, are specifically designed to handle high currents and store significant charges. The parallel configuration of the supercapacitors ensures that the energy is evenly distributed across the network, minimising the risk of overload and optimising energy storage.

The control unit (reference character 300) is critical in managing this process. During the energy storage phase, the control unit connects the antenna tower to the supercapacitors, allowing them to absorb and store the energy efficiently. This phase continues until the supercapacitors reach their maximum storage capacity.

Energy Management and Safety:

If, at any point, the incoming energy exceeds the storage capacity of the supercapacitors, the excess energy is diverted through a grounding mechanism (reference character 400). This mechanism, which includes Gas Discharge Tubes (GDTs) strategically placed at the end of each branch, directs the surplus energy safely to the ground, protecting the system from potential damage due to overloading.

Energy Discharge and Utilization:

Once the energy is stored, the control unit shifts from storage to discharge mode. In this phase, the stored energy within the supercapacitors can be released in a controlled manner. The control unit connects the supercapacitors to external loads or the electrical grid, enabling the harvested lightning energy to be used as a supplemental power source. This energy can be utilised during peak demand periods, power outages, or other scenarios where additional power is required.

Scalability and Adaptation:

The system is designed to be scalable and adaptable, allowing it to be integrated into various regional energy grids depending on specific energy needs and the frequency of lightning strikes in the area. The system's modular design facilitates easy expansion and customisation, making it suitable for various applications, from small-scale installations to large energy networks.

Fail-Safe Operations:

In extreme conditions or during particularly intense lightning storms, the system's multiple layers of protection, including the grounding mechanism and overvoltage protection circuits, work together to ensure the system's reliability and longevity. These safety features prevent system failure, protect the components, and ensure uninterrupted operation, even under the most demanding circumstances.

This method of operation allows the invention to harness the immense energy potential of lightning strikes safely and effectively, contributing to grid stability and renewable energy efforts. The seamless transition between energy capture, storage, and release ensures the system can provide a continuous and reliable power supply whenever needed.

The advantages and positive outcomes of the present invention include:

1. **Efficient Energy Conversion:** The invention offers a reliable method to directly capture and convert lightning energy into usable electrical power, addressing the inefficiencies and limitations of prior technologies.
2. **Enhanced Safety Mechanisms:** Integrating high-capacity Gas Discharge Tubes and multiple lightning arresters ensures the system's safety and longevity, even under extreme lightning strikes.
3. **High-Capacity Energy Storage:** Supercapacitors allow the system to store large amounts of energy, ensuring a steady power supply long after the lightning event.

This novel system offers a practical and efficient means of harnessing the vast energy potential of lightning, contributing to renewable energy efforts and grid stability.

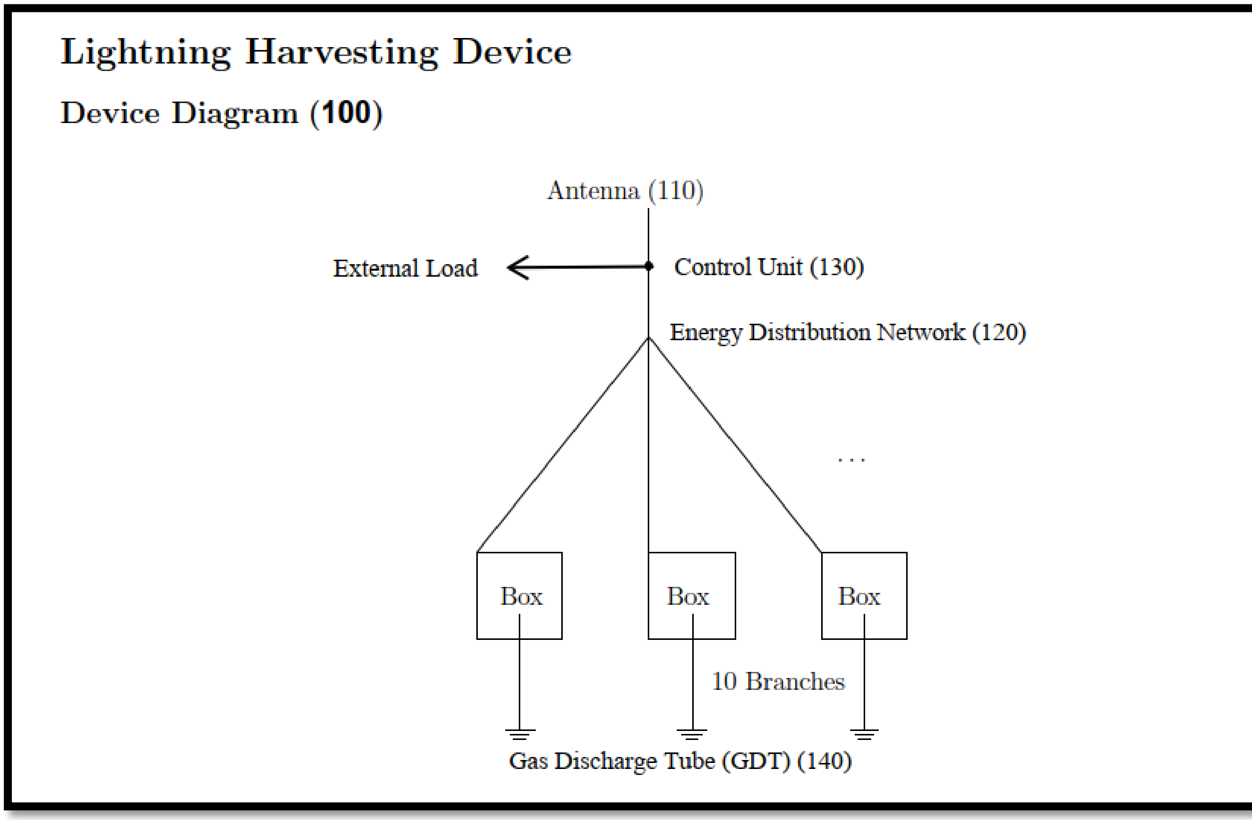

FIGURE 1

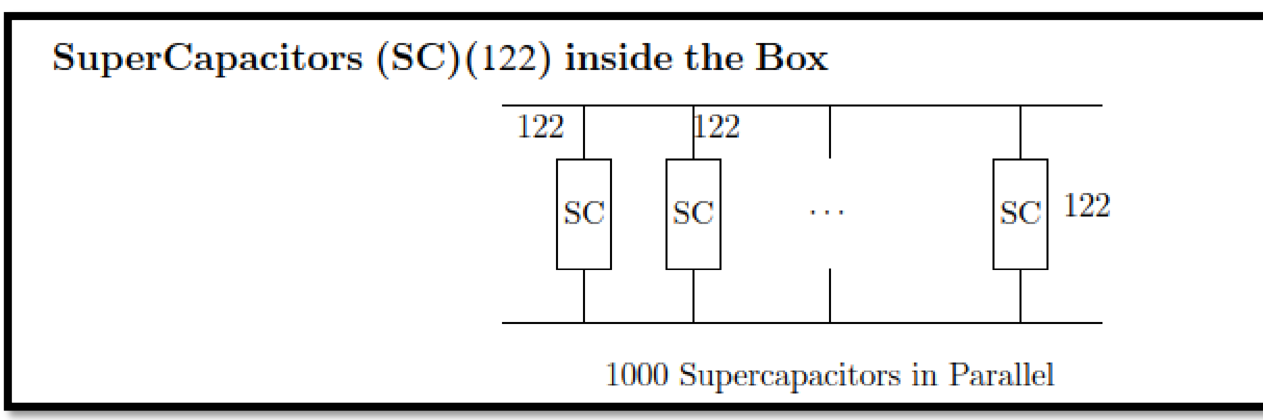

FIGURE 2

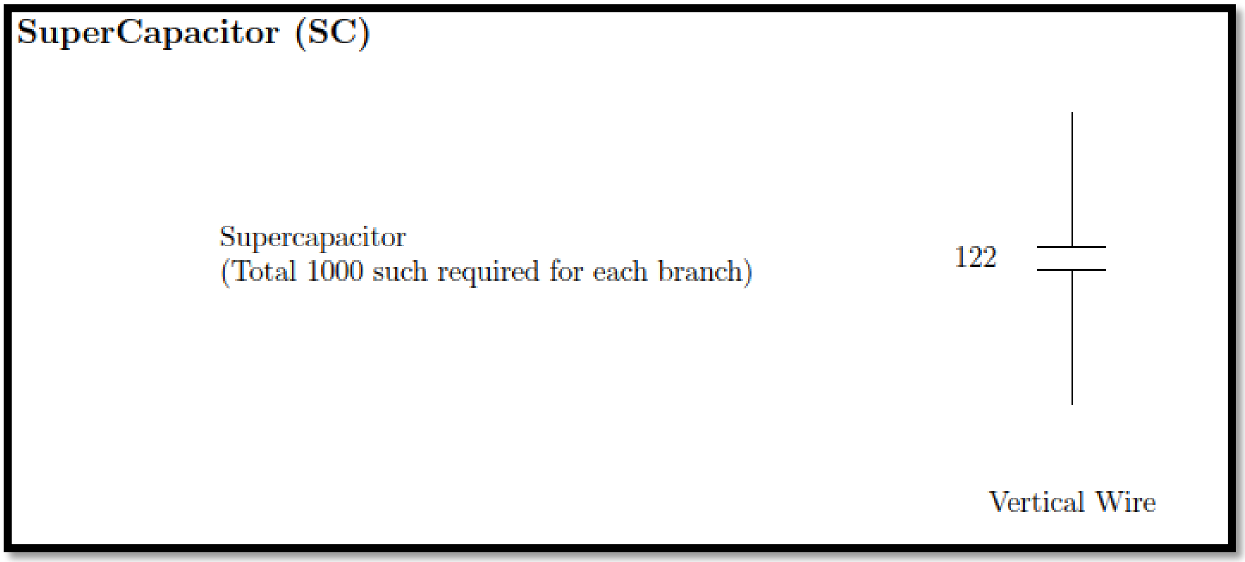
SuperCapacitor (SC)
Supercapacitor
(Total 1000 such required for each branch)
122
Vertical Wire

FIGURE 3

List of Reference Numbers

100	Lightning Harvester Device
110	Antenna
120	Energy Distribution Network
122	SuperCapacitors
130	Control Unit
140	Gas Discharge Tube

We claim:

1. A lightning harvesting system comprising:

    i. An omnidirectional conductive antenna tower designed to attract and capture lightning strikes;

    ii. an energy distribution network with ten branches, each containing high-capacity supercapacitors arranged in parallel, wherein the energy distribution network comprises 1,000 high-capacity supercapacitors per branch based on Graphene and/or Activated Carbon, with each supercapacitor capable of handling at least 100 amperes of current and storing at least 250 coulombs of charge at a voltage of 2.5 volts;

    iii. a control unit configured to manage the storage and release of the captured energy; configured to connect the lightning rod to the capacitors during the energy storage phase and to connect the capacitors to a load or further circuitry during the discharge phase;

    iv. a grounding mechanism comprising Gas Discharge Tubes (GDTs) strategically placed at the end of each branch within the energy distribution network, configured to direct excess current to the ground and handle up to 100,000 amperes of current from lightning strikes, ensuring safe discharge of excess voltage and protection of system components while distributing the captured energy across multiple branches for storage and later use.

2. The system of claim 1, wherein the conductive antenna tower is designed to maximise lightning strikes' capture area and efficiency.
3. The system of claim 1, wherein the conductive antenna tower includes materials and design features that optimise the capture of electrical energy from lightning strikes.
4. The system of claim 1, wherein the grounding mechanism ensures the safety and reliability of the system by preventing overload and directing excess current to the ground.

5. The system of claim 1, wherein the design is scalable to allow for broader implementation in various lightning-prone regions and integration into regional energy grids to supplement energy needs.
6. The system of claim 1, wherein the system provides additional power during peak demand periods or power outages, thereby enhancing grid stability and resilience.
7. The claim 1 system further comprises advanced materials such as Graphene or activated Carbon-based supercapacitors to improve the system's efficiency and capacity.
8. The system of claim 1, further comprising a control unit configured to manage the distribution and storage of energy captured from lightning strikes.

Dated this 18th day of August 2024.

ABSTRACT

SYSTEM FOR HARVESTING AND UTILIZING ELECTRICAL ENERGY FROM LIGHTNING STRIKES

The present invention is a lightning harvesting apparatus designed to capture and store electrical energy from lightning strikes. The apparatus includes an omnidirectional conductive antenna tower connected to an energy distribution network comprising multiple branches with supercapacitors arranged in parallel. The captured energy is directed into these supercapacitors for storage. A control unit is integrated to manage energy storage and release, while a grounding mechanism handles any excess current that exceeds the storage capacity. An overvoltage protection component also safeguards the system from extreme voltages, ensuring operational safety. With the above-described structure, the apparatus efficiently converts the high-voltage energy from lightning into a manageable DC power supply, which can be utilised in various applications. The system is designed for scalability and integration into existing energy grids, providing supplemental power and enhancing grid stability.

Figure 1, 2 and 3 are the representative figures.